Globalization and Economic Statecraft

US–China AI competition: networks, institutions and responses

PSCI 410 | Week 9 | 16 November 2026 | Prof. Hongying Wang; guest Hao Xi

Before class: read Farrell and Newman, ch. 2, and Goddard, ch. 4, in Drezner, Farrell and Newman, eds., *The Uses and Abuses of Weaponized Interdependence* (2021), and this page. No other preparation.

Central question: How does global interdependence create—and limit—economic leverage in US–China AI competition?

By the end of the session, you should be able to:

1. Explain how global networks create hubs that states can turn into weapons, using AI chips.
2. Analyse how domestic institutions and firms shape the use of these weapons.
3. Assess China's efforts to reduce dependence and gain leverage.

Background 1: how AI models are trained. An AI model learns by running an enormous number of calculations over large amounts of data on specialised chips. DeepSeek reported that its V3 model needed 2.788 million H800 chip-hours to train: about 320 years for a single chip, done on a cluster of 2,048 chips, with pre-training in under two months. More and better chips mean faster training and larger models. That is why chips are the part of AI that governments can count, trace and restrict.

Background 2: how an AI chip is made. No single country makes one.

1. Design	**2. Manufacturing**	**3. Assembly**	**4. Use**
NVIDIA (US)	TSMC (Taiwan) or Samsung (Korea), on lithography machines from ASML (Netherlands)	Testing and packaging firms	AI developers in China and elsewhere

Each step sits in a different jurisdiction. Owning a step and having the legal power to restrict it are different things.

Background 3: the case. In October 2022 the United States restricted exports of advanced AI chips and chip-making items to China, citing national security. NVIDIA redesigned chips to fall outside the rules, and the chip industry asked the US export agency (BIS) for simpler procedures. Chinese firms built domestic chips; in 2025 they shipped about 7% of NVIDIA's volume. In 2025–26, Washington licensed some NVIDIA chips for China, but Beijing restricted the sales.

Readings as tools. *Farrell and Newman:* a state can turn a network position into a weapon only with legal jurisdiction and institutions to use it; targets respond. *Goddard:* challengers can gain influence by building international ties, such as a competing hub.

What you will do in class

- **Opening:** scan the QR code below, or use paper, to make a private prediction.
- **Your perspective:** you will count off into A (US policy), B (Chinese policy) or C (chip companies). During the lecture you write short notes from that perspective on your worksheet. No one is asked to raise a hand.
- **Discussion panel (minute 50):** join your letter's group and build your opening from your notes: *Do US chip restrictions preserve American leverage, or encourage changes that weaken it over time?* Analyse the perspective; you need not defend it.
- **For Week 11:** borrow one device from today (map, timeline or document comparison) for your findings presentation.

Case page: `https://erwinpsci.github.io/psci410-ai-statecraft-teaching/`
No login; choices stay on your device and nothing is collected. Paper works equally well. No student GenAI use is required.

Sources: DeepSeek-V3 technical report (2024); NVIDIA FY2023 annual report and Form 10-Q (2026); ASML; BIS rules (2022, 2024); SIA comments (2024); The Substrate (2 June 2026). Links appear on the case page and in the slides' source lines.